
An LLM Judge Loses Reliability on Problems in Its Training Corpus, but Only When the Reference Answer Is Withheld

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Abstract

Automated evaluators are trained artifacts, so a verdict on a benchmark item may depend on whether the evaluator read that item during training, yet the corpora of models deployed as judges are closed and the question is usually left open. Instella publishes its complete pretraining mixture, and its stage two component derives from the GSM8K training split and not the test split, so membership is fixed by exact thirteen gram containment rather than estimated: the adopted threshold flags none of 1000 known negatives and the looser any match rule flags 3, a false positive rate of 0.3 percent. Difficulty matched arms of 250 seen and 250 unseen problems expand into 2017 candidate solutions from four Instella checkpoints, and five judges spanning three model families and 3 to 32 billion parameters grade them at temperature zero under two prompts differing only in whether the gold answer is supplied. Chance corrected agreement is reported as a cautionary result rather than a metric: Cohen’s kappa manufactures a seen minus unseen gap of -0.292 [$-0.356, -0.228$] from a base rate difference of 77 against 53 percent, a textbook instance of the Feinstein and Cicchetti paradox, where balanced accuracy on identical verdicts gives $+0.054$. Under balanced accuracy every judge shows the same dissociation. With the reference supplied the membership gap stays between -0.02 and -0.07 ; withholding it moves every judge to between -0.08 and -0.24 , a within judge ratio of 1.5 to 8.0, and the gap is present at the 32 billion parameter judge whose reference free accuracy is the highest measured. Decomposing it locates the effect precisely. Sensitivity is unchanged by membership, $+0.011$ [$-0.005, +0.034$] and -0.004 [$-0.024, +0.016$], while specificity collapses, -0.496 [$-0.573, -0.415$] and -0.378 [$-0.472, -0.276$]: one judge rejects 52 percent of wrong solutions to unseen problems and 3 percent of wrong solutions to seen ones. Because the same solutions are graded under both prompts, the control separates this from the possibility that errors on seen problems are simply subtler, which accounts for about a third of the gap and no more. On a familiar problem without a reference these judges become credulous rather than confused, losing the ability to reject a wrong answer while retaining the ability to accept a right one, and the pattern survives a reworded prompt. Reference free grading is the configuration used wherever no gold answer exists, and it is least reliable on exactly the items most likely to sit in a judge’s training data.

1 Introduction

Using a language model to grade another model’s output rests on an assumption that is seldom examined, that the evaluator’s reliability does not depend on the material being evaluated. If a judge has memorized a benchmark problem, its verdict on a candidate solution to that problem may reflect

37 what it remembers instead of what it was shown. The verdict is then a measurement whose calibration
38 varies with the thing being measured, which is the property an instrument is supposed not to have.

39 Testing the assumption needs the judge’s training corpus. For the closed weight models usually
40 deployed as judges that is unavailable, so existing audits study properties observable from the outside.
41 Zheng et al. [Zheng et al., 2023] establish the paradigm and document position bias, verbosity bias
42 and self preference, and report that strong judges agree with human preference above 80 percent of
43 the time. None of that work can condition on membership, because membership is not knowable for
44 the judges in question.

45 Instella [Liu et al., 2025] makes it knowable for one judge and one corpus. AMD publishes the
46 full pretraining mixture, and the stage two synthetic mathematics corpus derives from the GSM8K
47 training split [Cobbe et al., 2021] and not from the test split. Membership of a problem in that corpus
48 follows from exact thirteen gram containment [Brown et al., 2020]. Since the corpus is generated only
49 from train, every test item is a known negative, which converts detector error from an assumption
50 into a measured quantity.

51 That gives an item partition with verified provenance, and the study puts the same items to three
52 judges under two prompts differing in one respect, whether the gold answer for the graded variant
53 appears. The reference based condition reduces grading to comparison, so a remembered answer
54 buys the judge nothing. The reference free condition requires assessing the presented reasoning,
55 which is where recall could substitute for verification. Because both conditions run over identical
56 items and identical candidate solutions, the contrast between them is internal to each judge and does
57 not depend on what the arm label means for that judge’s own corpus, a point Section 6 returns to.

58 Three contributions follow. First, judge reliability measured against membership fixed by containment
59 rather than inferred, over 48,408 verdicts. Second, a dissociation in which supplying a reference
60 answer removes the membership gap while withholding it does not, with the gap appearing precisely
61 where judges become competent. Third, a documented case of the kappa paradox producing a large
62 and interval excluding effect on data where the effect is absent, which is worth reporting because
63 chance corrected agreement remains a common choice for exactly this measurement.

64 2 Design

65 **Verified arms.** Item text is lowercased, reduced to alphanumeric tokens, and cut into every window
66 of 13 consecutive tokens. Containment is the fraction of an item’s windows appearing in the corpus.
67 Because the corpus derives only from GSM8K train, the test split calibrates the rule: the any match
68 convention of Brown et al. [Brown et al., 2020] flags 3 of 1000 known negatives, and the threshold
69 adopted here flags none of them. Identifiers are namespaced by split, without which train and test
70 items collide and containment exceeds unity. From the eligible pools, 250 seen and 250 unseen
71 parent problems are drawn matched on reasoning step count, so arm composition cannot stand in for
72 a membership effect.

73 **Candidate solutions.** Each parent expands into one original and four variants, two answer pre-
74 serving and two answer changing, giving 2017 rows. Four Instella checkpoints generate candidate
75 solutions under pinned decoding, and exact match against ground truth supplies the label each judge
76 is scored against. Two thirds of graded rows are variants, and the answer changing variants carry
77 different correct answers from their originals, which matters for the mechanism in Section 5.

78 **Judges and conditions.** Three judges span a deliberate capability range: Instella 3B Instruct,
79 Qwen2.5 7B Instruct and Qwen2.5 14B Instruct. Each emits one word, correct or incorrect, at
80 temperature zero under a 24 token budget. The two conditions differ in one element of the prompt.
81 Reference based grading supplies the gold answer for the variant at hand and is the control, since
82 recall of a remembered solution cannot help once the solution is given. Reference free grading
83 withholds it.

84 **Metric and inference.** Balanced accuracy, the unweighted mean of sensitivity and specificity, is
85 used throughout. Both components condition on the true class, so neither moves with the base rate,
86 and their mean stays comparable across arms whose ground truth composition differs. Chance is 0.5.
87 Section 3.1 shows why the obvious alternative fails here. Intervals are bootstraps over parent problems

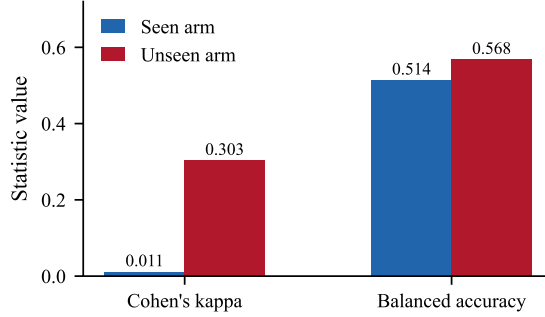


Figure 1: Two statistics computed on one set of verdicts, Instella 3B Instruct grading stage two outputs in the reference free condition. Cohen’s kappa reports a large seen minus unseen gap that reflects a ground truth base rate difference of 77 against 53 percent between the arms. Balanced accuracy, which conditions on the true class, does not.

rather than rows, because the five rows of one problem are not five independent observations; the measured intraclass correlation is approximately 0.48, and resampling rows would contract intervals by a design effect near 2.4.

3 Results

3.1 Chance corrected agreement is the wrong statistic here

Judge accuracy was first measured as Cohen’s kappa against exact match ground truth, and the result looked conclusive. Grading stage two outputs, the seen minus unseen difference came to -0.292 on an interval of $[-0.356, -0.228]$, and grading instruct outputs -0.168 on $[-0.228, -0.113]$. Both exclude zero by a wide margin and both point the same way.

The effect is an artifact of the marginals. Kappa depends on the marginal distributions as well as on agreement, and the arms have very different ground truth base rates: the instruct generator is correct on 77 percent of seen rows against 53 percent of unseen. An arm sitting near one half has far more chance corrected headroom than a skewed arm, so the skew alone produces a gap. Feinstein and Cicchetti [Feinstein and Cicchetti, 1990] describe this as the first of two kappa paradoxes, in which high raw agreement is driven down by marginal imbalance. Figure 1 puts both statistics on the same verdicts. Kappa separates the arms by 0.292; balanced accuracy separates them by 0.054 in the opposite direction.

Reporting this is not incidental. Chance corrected agreement is a standard choice for judge reliability, and on a design where the arms differ in difficulty it will manufacture a membership effect from nothing.

3.2 The dissociation

Table 1 and Figure 2 give the membership gap in balanced accuracy under both conditions, for five judges spanning three model families and a capability range from 3 to 32 billion parameters. The control behaves as a control: with the reference supplied the gap stays small everywhere, between -0.02 and -0.07 . Withholding it moves every judge to between -0.08 and -0.24 , and the within judge ratio of the two runs from 1.5 to 8.0. That ratio, rather than any trend across the ladder, is what the design identifies, since it compares one judge against itself on identical items and identical candidate solutions.

Scale does not remove the reference free gap. It is present at every judge tested, including the 32 billion parameter one, whose reference free balanced accuracy on the seen arm is the highest measured at 0.648 and whose gap is still -0.191 . A defect of weak models would shrink with capability; this does not.

Table 1: Balanced accuracy, seen minus unseen, grading instruct outputs, across five judges from three model families. Intervals are 95 percent bootstraps over parent problems, 4000 draws; an asterisk marks an interval excluding zero. The ratio is the reference free gap divided by the control gap for that judge.

Judge	Reference given (control)	Reference withheld	Ratio
Instella 3B Instruct	-0.054 [-0.083, -0.026]*	-0.081 [-0.109, -0.054]*	1.5
Llama 3.1 8B Instruct	-0.074 [-0.116, -0.032]*	-0.243 [-0.281, -0.201]*	3.3
Qwen2.5 7B Instruct	-0.036 [-0.068, -0.008]*	-0.200 [-0.248, -0.149]*	5.6
Qwen2.5 14B Instruct	-0.023 [-0.057, +0.006]	-0.184 [-0.231, -0.133]*	8.0
Qwen2.5 32B Instruct	-0.062 [-0.103, -0.021]*	-0.191 [-0.239, -0.140]*	3.1

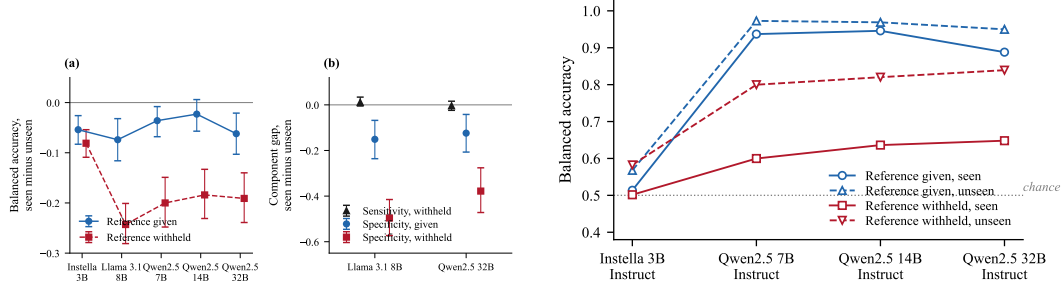


Figure 2: Left: membership gap in balanced accuracy across the judge ladder under both conditions, grading instruct outputs. Right: absolute balanced accuracy for all four cells of the design across the capability ladder. The control condition confirms the three larger judges are competent instruments, reaching 0.89 to 0.97 on the seen arm, so the reference free result is not read off a judge that cannot grade. The two control series converge as capability rises while the two reference free series stay apart.

3.3 Capability is demonstrated rather than assumed

A null or an effect from an incompetent judge says nothing, so absolute performance is reported alongside the gaps. Table 2 gives every cell. Given the reference answer, the 7, 14 and 32 billion parameter judges reach 0.888 to 0.973, which is the regime in which their reference free numbers become interpretable. The 3 billion parameter judge sits within 0.04 of chance on the seen arm in every configuration, including when handed the answer, so its rows describe a noise floor and the substantive claims rest on the larger three.

Three readings follow from the table taken together with Table 1. Capability is real, so conclusions about the instrument come from instruments that work. The control behaves as a control, since supplying the correct answer for the specific variant removes the gap, which is what a recall account predicts and a pure difficulty account does not. And the reference free gap neither converges nor responds to scale.

3.4 Abstention and what would falsify the account

A judge that declines to answer more often on one arm could produce the whole pattern with no change in grading quality at all, since abstentions removed from the denominator reshape both sensitivity and specificity. Abstention stays between 0.0 and 2.6 percent across every judge, condition and arm, which is too small and too evenly spread to carry a gap of 0.200.

Three outcomes would have contradicted the account, and none occurred. Had the reference free gap shrunk with capability in the way the control gap does, the effect would be a defect of weak judges; instead it grows from the first judge to the second and then holds. Had the control gap failed to converge, the arms would differ in some way that persists even when the answer is supplied, pointing at difficulty rather than at recall; instead the strongest judge's control interval spans zero. And had the two larger judges failed to clear chance in the control condition, no reading of their reference free numbers would be available at all; instead they reach 0.937 and 0.946 on the seen arm.

Table 2: Absolute balanced accuracy per cell, grading instruct outputs. Chance is 0.500.

Judge	Reference given		Reference withheld	
	Seen	Unseen	Seen	Unseen
Instella 3B Instruct	0.514	0.568	0.502	0.583
Llama 3.1 8B Instruct	0.860	0.934	0.511	0.754
Qwen2.5 7B Instruct	0.937	0.973	0.600	0.800
Qwen2.5 14B Instruct	0.946	0.969	0.636	0.820
Qwen2.5 32B Instruct	0.951	0.968	0.648	0.839

4 Implications for evaluator construct validity

An evaluator is valid for a construct when its verdicts track that construct and nothing else. The results here describe a specific and narrow failure of that condition. Under reference free grading the measured quantity is not solely the quality of the reasoning presented, because it also varies with whether the graded problem is one the evaluator is likely to have read. The variation is not small: 0.200 in balanced accuracy on the 7 billion parameter judge, against a control gap of 0.036 on the identical items and the identical candidate solutions.

Two practical consequences follow. Reference free grading is the configuration used exactly where a gold answer does not exist, which includes open ended generation, proof sketches, and any setting where the task is to assess a chain of reasoning on its own terms, so the failure falls on the applications that most need a trustworthy judge. And a benchmark assembled from public sources is the case where the affected items are most concentrated, since public availability is what puts an item in a training corpus in the first place. Supplying a reference answer, where one exists, removes the effect in these data, which makes it a mitigation rather than merely a diagnosis.

4.1 Robustness to the prompt

The dissociation is not an artifact of one wording. Repeating both judges under a reworded prompt carrying the same information, the reference free gap is -0.254 $[-0.292, -0.212]$ for Llama 3.1 8B against -0.243 under the original, and -0.218 $[-0.262, -0.170]$ for Qwen2.5 32B against -0.191 , with the control gaps staying small at -0.051 and -0.086 .

5 Where the gap comes from

Balanced accuracy has two components and they behave differently, which narrows what the effect can be. Sensitivity, the rate at which genuinely correct solutions are accepted, is unchanged by membership: $+0.011$ $[-0.005, +0.034]$ for Llama 3.1 8B and -0.004 $[-0.024, +0.016]$ for Qwen2.5 32B. The entire gap is specificity, the rate at which genuinely wrong solutions are rejected, at -0.496 $[-0.573, -0.415]$ and -0.378 $[-0.472, -0.276]$. Llama rejects 52.3 percent of wrong solutions to unseen problems and 2.7 percent of wrong solutions to seen ones.

This rules out the reading that first suggested itself, that the judge compares a candidate against a remembered answer instead of checking it. That account predicts correct solutions being marked wrong where a variant changed the answer, and therefore a sensitivity loss. Sensitivity does not move, and a direct test on the answer changing variants, where a remembered answer would be the wrong answer, finds no concentration of false negatives on the seen arm. What the judge does on a familiar problem is accept, not misremember.

One confound has to be separated before that means anything about membership. The generator is more accurate on seen problems, so its errors there may simply be subtler and harder to catch, which would depress specificity with no involvement of the judge’s own training data. The control settles it, because both conditions grade the same solutions. With the reference supplied the specificity gap is -0.151 $[-0.236, -0.068]$ and -0.124 $[-0.207, -0.042]$, so part of the effect is indeed error subtlety. Withholding the reference roughly triples it. That excess cannot be a property of the solutions, since the solutions are identical.

The residual account is narrower than a memorisation story and easier to act on. Asked to judge a familiar problem without a reference, these judges become credulous rather than confused: they lose

185 the ability to reject a wrong answer while retaining the ability to accept a right one. Supplying a
186 reference restores most of it.

187 6 Limitations

188 **Membership is verified for one judge, and it is not the judge carrying the result.** Containment
189 is measured against the Instella corpus, so the Instella judge has verified membership but sits at
190 chance. The Qwen judges show the effect and their corpora are unknown, so for them the partition
191 is GSM8K train against test rather than verified seen against unseen. That asymmetry is why the
192 primary claim is stated as a within judge contrast between two conditions over identical items and
193 identical candidate solutions, which holds whatever the arm label means for a given judge. Closing
194 the gap to a membership claim requires a second open corpus model of comparable capability.

195 **One family supplies the candidate solutions.** Every graded output comes from an Instella check-
196 point, and whether the dissociation reproduces on other generators is untested. The judges span
197 Instella, Llama and Qwen.

198 **The effect depends on how gradeable the solutions are.** Grading the weaker stage two generator,
199 Qwen2.5 32B shows a reference free gap of -0.035 $[-0.075, +0.006]$, which spans zero, while
200 Llama 3.1 8B on the same target shows -0.192 . Blunt errors are catchable without a reference,
201 and reference free specificity is correspondingly high. The ceiling in solution quality that Section 5
202 predicts is therefore measured on one generator pair and is an observation rather than a law.

203 **One checkpoint is unexplained.** Stage one outputs are wrong nearly everywhere and trivially
204 gradeable in either condition, which accounts for that null. The supervised fine tuned checkpoint
205 shows no gap in either condition despite output quality comparable to instruct, and the present data
206 do not explain why.

207 **Attribution is not established.** No evidence links a verdict to a specific training document. The
208 decomposition in Section 5 narrows the effect to a specificity loss surviving the error subtlety control,
209 which is not attribution.

210 7 Conclusion

211 Measured against membership fixed by containment rather than estimated, an LLM judge shows
212 no loss of reliability on problems from its training corpus when the reference answer is supplied,
213 and a substantial one when it is withheld. The gap under reference free grading runs to 0.200 in
214 balanced accuracy and does not close with scale, while the same judges in the control condition
215 converge toward zero as they improve. Chance corrected agreement inverts this picture on identical
216 verdicts and should not be used where the arms differ in base rate. The operational reading is specific:
217 reference free grading is the configuration deployed whenever no gold answer exists, and it is the
218 configuration in which the items most likely to be in a judge’s training data are graded least reliably.

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253 **A Related work in full**

254 Zheng et al. [Zheng et al., 2023] introduce the LLM as judge paradigm at scale and document position
255 bias, verbosity bias and self preference, reporting agreement with human preference above 80 percent
256 for strong judges. Their audits characterize the judge and the prompt; membership of the graded item
257 in the judge’s corpus is outside reach for the closed models they study, which is the gap the present
258 design addresses for one open family.

259 On the detection side, containment based overlap rules originate with Brown et al. [Brown et al.,
260 2020], and membership inference from model confidence is developed by Carlini et al. [Carlini et al.,
261 2023] and Shi et al. [Shi et al., 2024]. Yang et al. [Yang et al., 2023] show that rephrasing defeats
262 overlap based detection outright, which is why a study that needs a trustworthy treatment label cannot
263 rely on an estimated one.

264 Two lines of work establish the value of released intermediate checkpoints. Biderman et al. [Biderman
265 et al., 2023b] release the Pythia suite specifically to make training dynamics analyzable, and Biderman
266 et al. [Biderman et al., 2023a] use such checkpoints to study when memorization emerges during
267 training. The present work applies the same logic to an evaluator rather than to a generator.

268 **B Containment method and calibration**

269 Item text is normalized to lowercase alphanumeric tokens and decomposed into every window of 13
270 consecutive tokens. Containment against a corpus is the fraction of an item’s windows found in it,
271 computed in one streaming pass with an inverted index from window to source item, so cost is linear
272 in corpus size and independent of how many item sets are indexed together. Because the corpus is
273 generated only from the GSM8K training split, every test item is a verified negative: the any match
274 rule flags 3 of 1000, a rate of 0.3 percent, and the adopted threshold flags none.

275 Identifier namespacing is load bearing. GSM8K train and test share an index scheme, so loading
276 both without namespacing collides items and produces containment values above unity, which is
277 impossible for a fraction. Anyone reproducing this design will meet the same trap.

278 **C Reproducibility controls**

279 Several controls are load bearing rather than incidental. The transformers version is pinned exactly,
280 since a version range places different checkpoints on different minor versions and that difference
281 enters the estimate as though it were a model difference. Batch size is fixed across all generation,
282 because greedy decoding in reduced precision is sensitive to padding and a batch size change alters
283 tie breaking. Variants are seeded per pair of item and perturbation type, so variant text is stable
284 regardless of call order and generations stay valid when the item set changes. Every generation batch
285 is flushed to disk and resume is keyed on item identifier, so an interrupted run continues instead of
286 restarting.

287 All generation records, containment values, judge verdicts and analysis code are released for audit at
288 a public repository and dataset mirror, both anonymized for review. Random seeds are fixed across
289 arm selection, variant construction and every bootstrap. An assistant helped implement generation,
290 containment and analysis code under the direction of the authors, who verified every reported number
291 against the underlying verdict files; the assistant did not design the conditions or choose which results
292 to report.